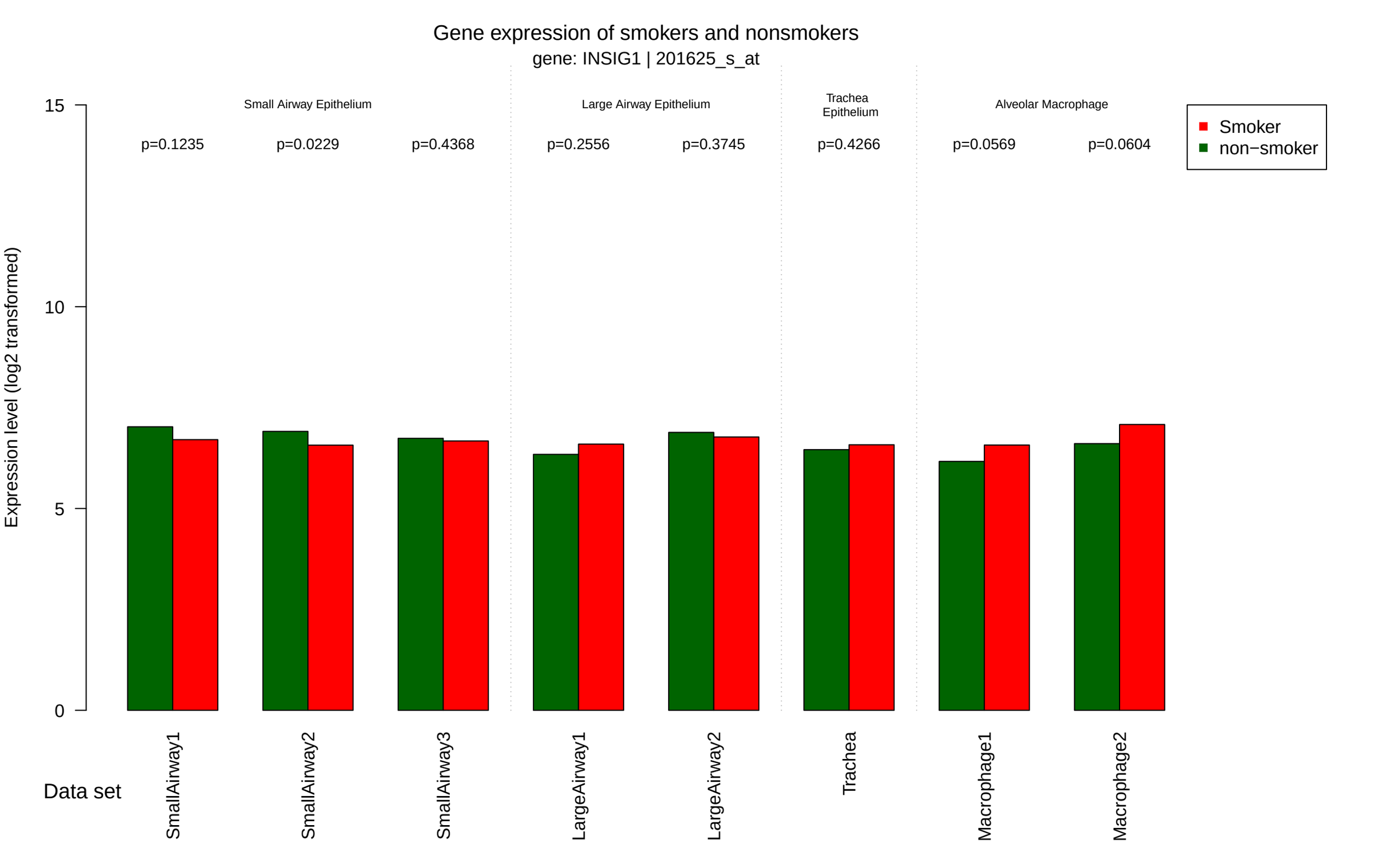
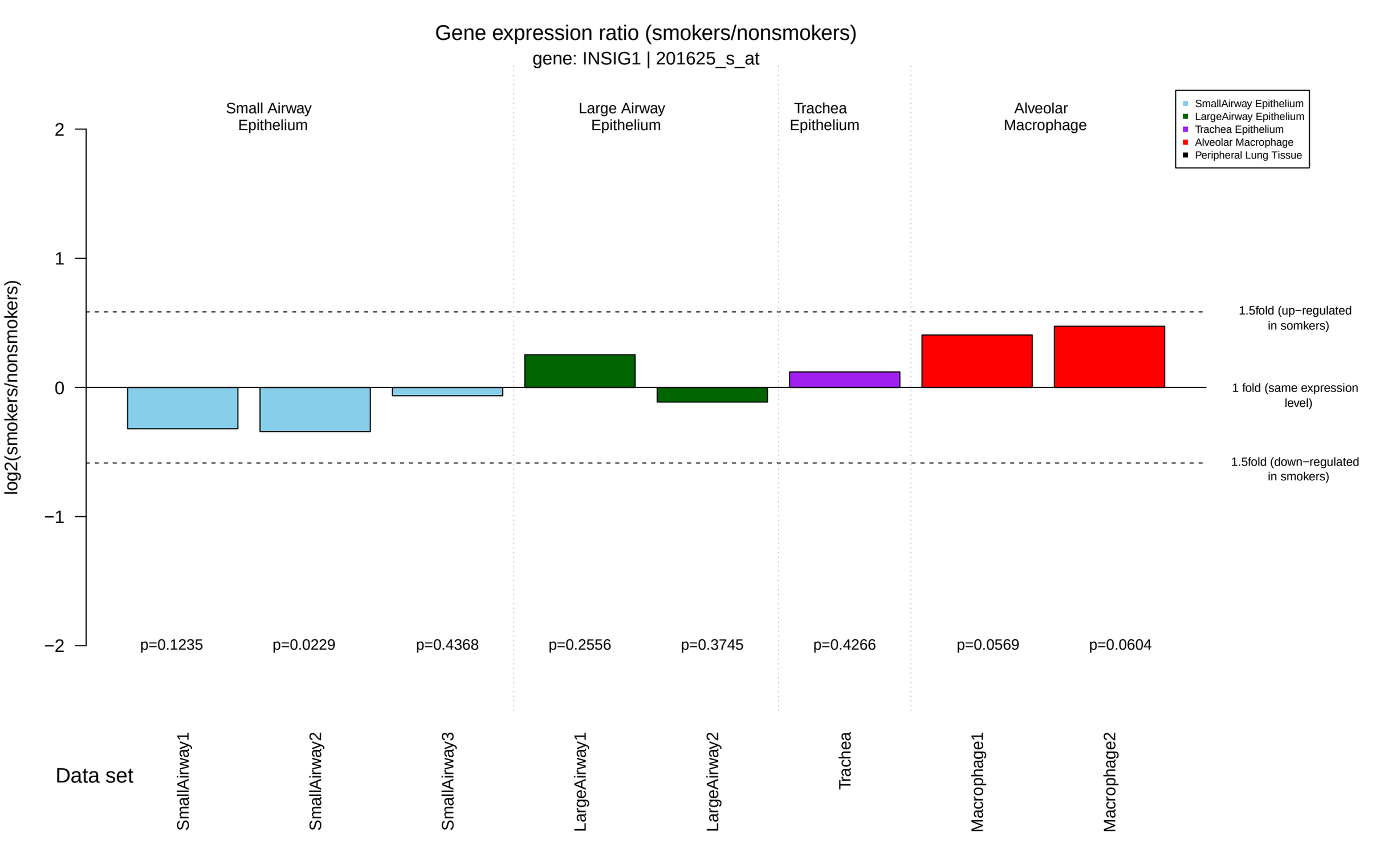
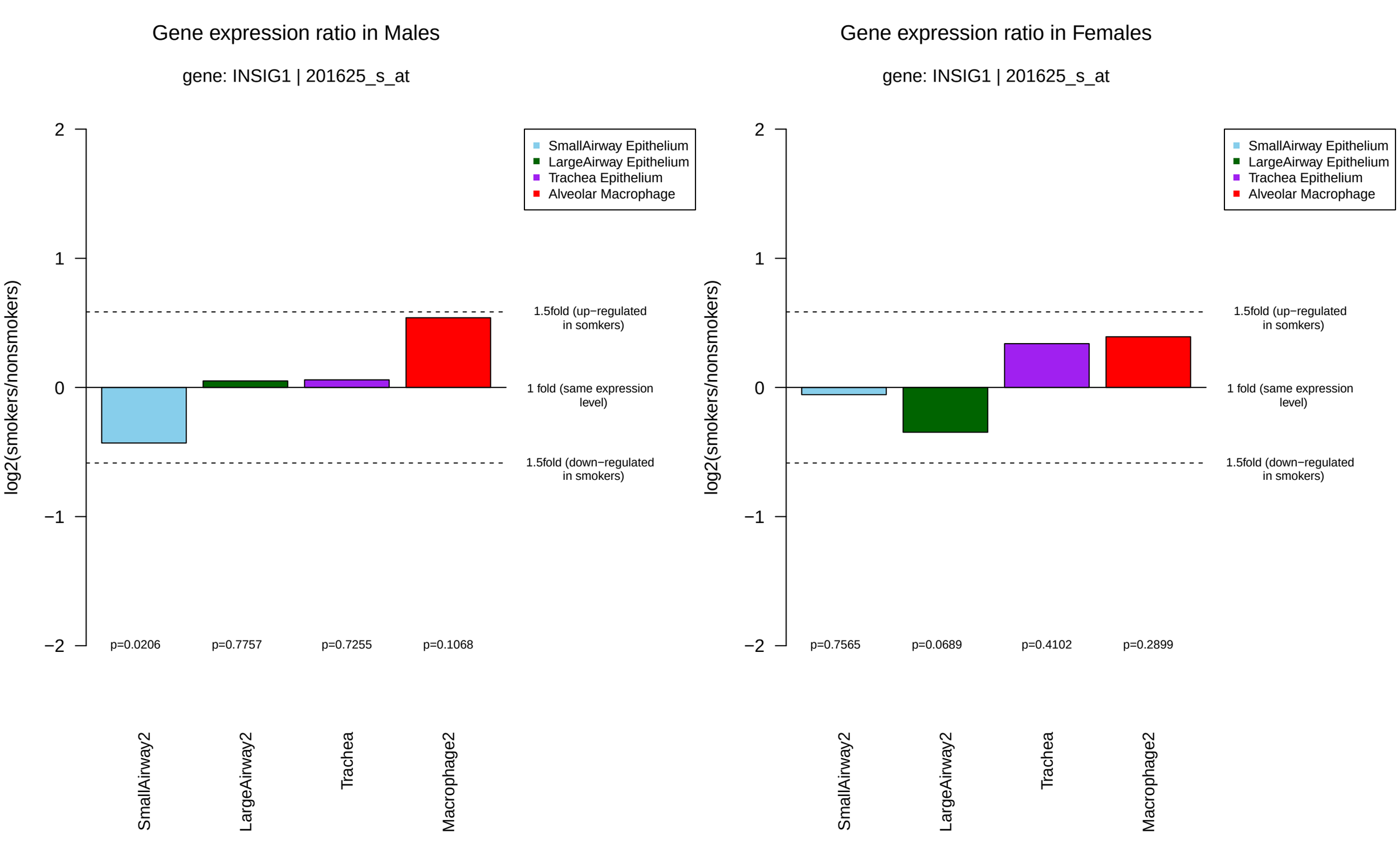
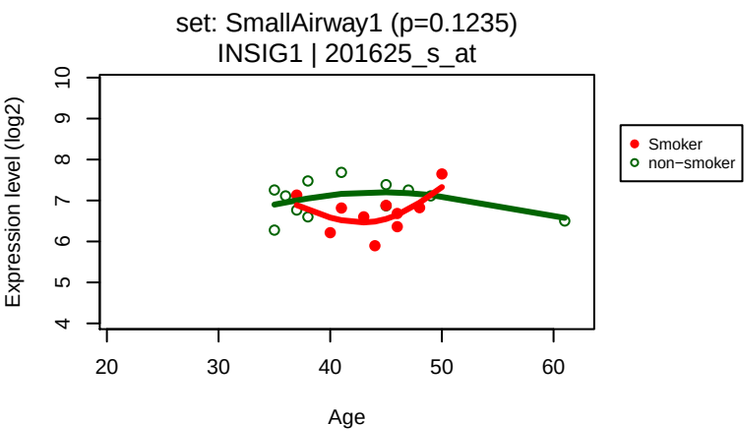
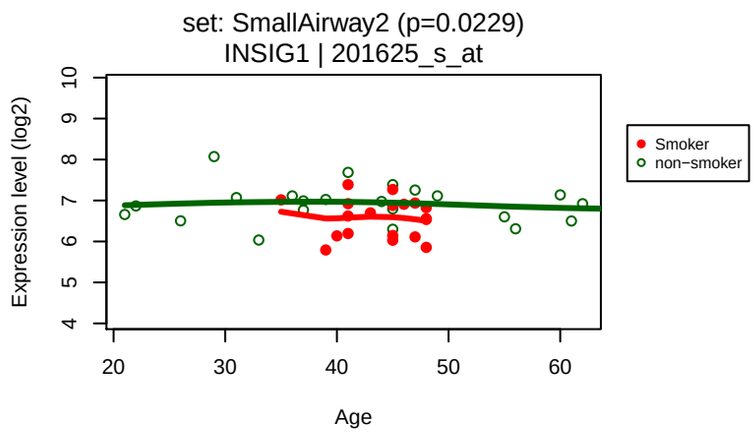
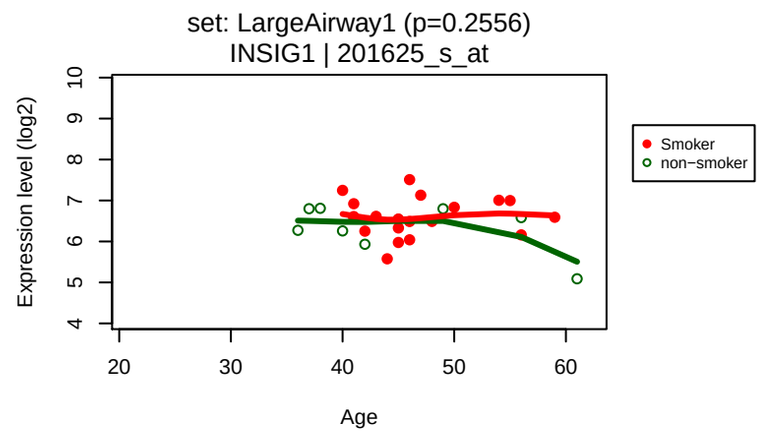

Gene expression of smokers and nonsmokers
gene: INSIG1 | 201625_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1235
p=0.0229
p=0.4368
p=0.2556
p=0.3745
p=0.4266
p=0.0569
p=0.0604
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

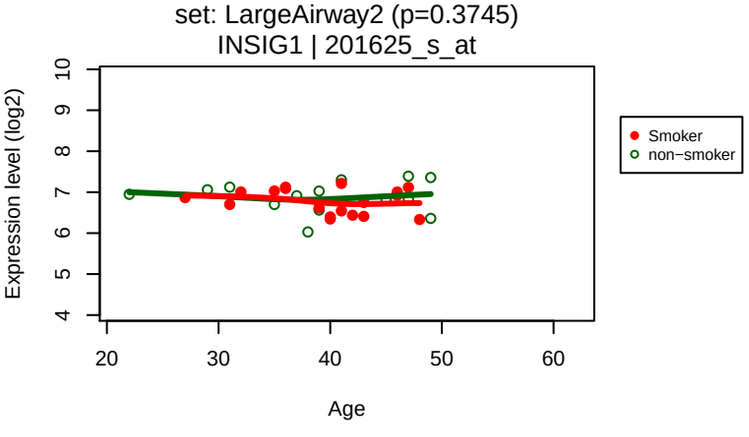

Gene expression ratio (smokers/nonsmokers)
gene: INSIG1 | 201625_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1235
p=0.0229
p=0.4368
p=0.2556
p=0.3745
p=0.4266
p=0.0569
p=0.0604
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

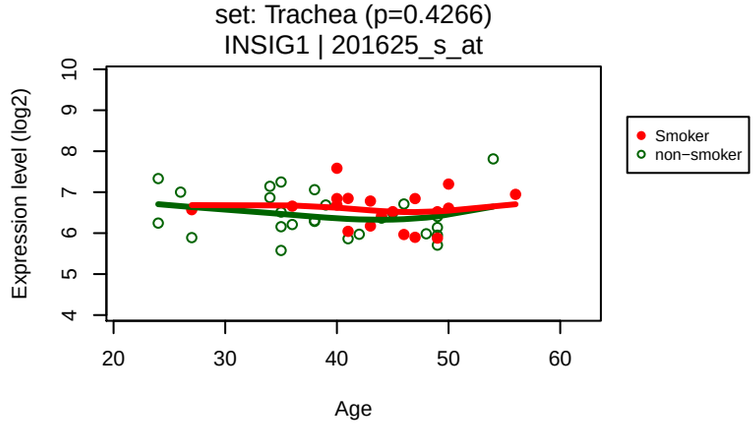

Gene expression ratio in Males
gene: INSIG1 | 201625_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0206
p=0.7757
p=0.7255
p=0.1068
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: INSIG1 | 201625_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7565
p=0.0689
p=0.4102
p=0.2899
SmallAirway2
LargeAirway2
Trachea
Macrophage2

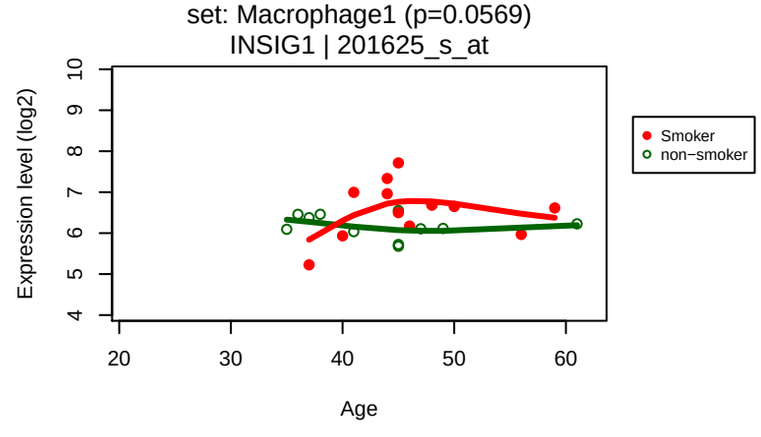

set: SmallAirway1 (p=0.1235)
INSIG1 | 201625_s_at
Expression level (log2)
Age
Smoker
non−smoker

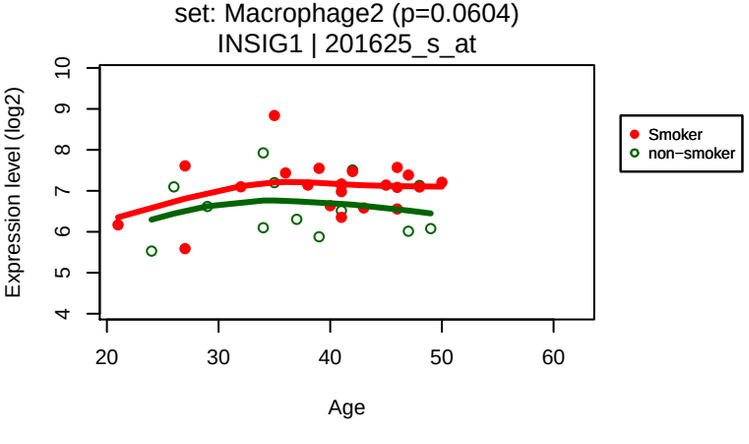

set: SmallAirway2 (p=0.0229)
INSIG1 | 201625_s_at
Expression level (log2)
Age
Smoker
non−smoker

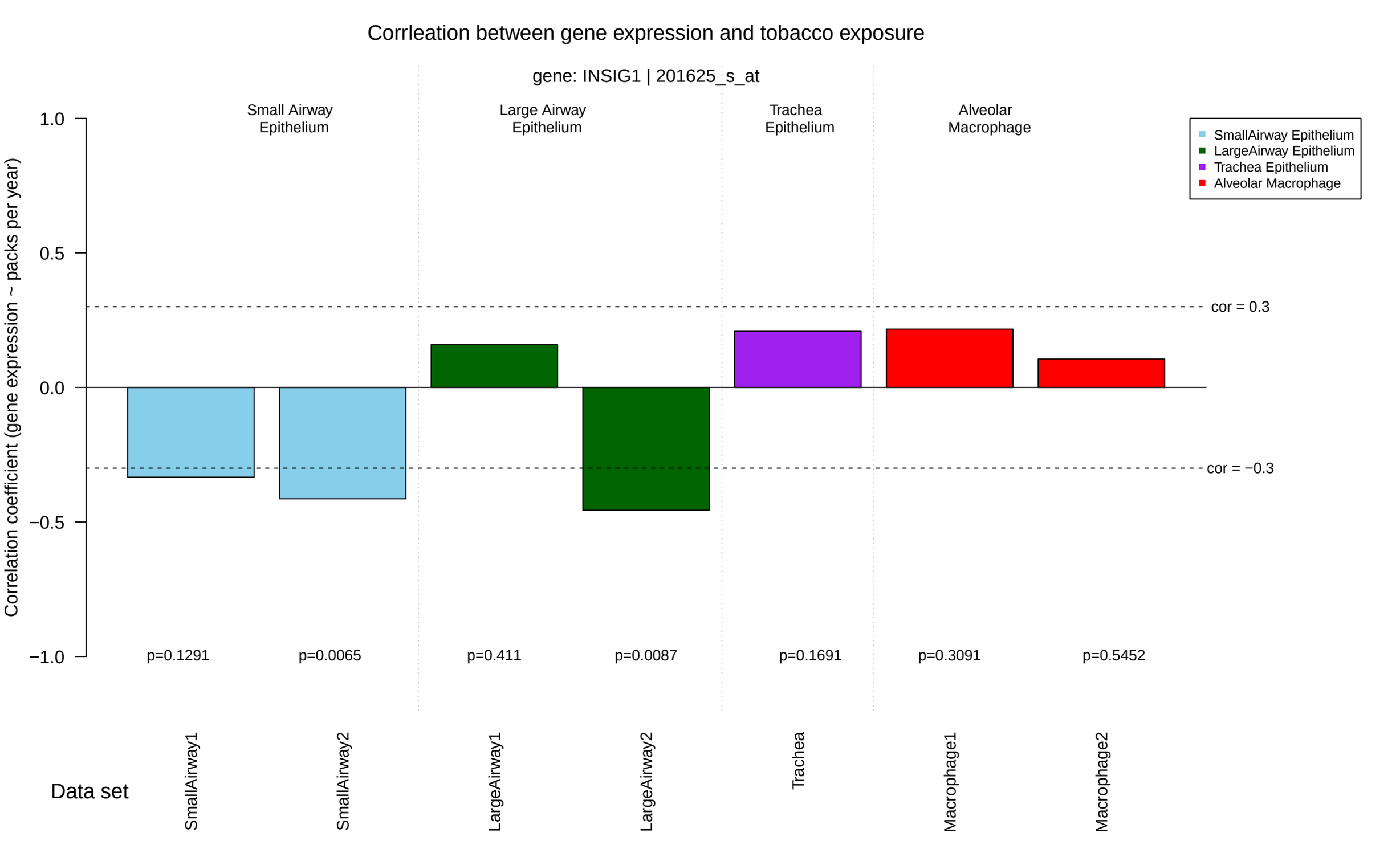

set: LargeAirway1 (p=0.2556)
INSIG1 | 201625_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.3745)
INSIG1 | 201625_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.4266)
INSIG1 | 201625_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0569)
INSIG1 | 201625_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0604)
INSIG1 | 201625_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: INSIG1 | 201625_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1291
p=0.0065
p=0.411
p=0.0087
p=0.1691
p=0.3091
p=0.5452
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2